National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Fundamentals of Databases	Course Code:	CS2011
	Program:	BS Electrical Engineering	Semester:	Fall 2026
	Duration:	10 minutes	Total Marks:	10
	Quiz Date:	08-Sep-2026	Weight:	
	Section:	BEE-7C	Page(s):	1
	Exam Type:	Quiz- 1	CLO #	1

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Student Nam[redacted] Roll No:[redacted]

Question No. 1 (CLO 1, PLO 1, C2) **Marks: 10**

i. **Explain** what is meant by logical data independence in 3-schema architecture of DBMS.

ii. **Express** the difference between data and metadata with an example.

iii. Consider the following relations for a database that keeps track of business trips of salespersons in a sales office:

CUSTOMER(CNIC, Name) PRODUCT(pid, name, price, company)

ORDER(oid, odate, cid) ORDER_DETAIL(oid, pid, quantity)

Determine and deny the foreign keys for this schema

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iv. Consider the following state of the schema above and **determine** if the operations listed below will pass or fail. If they will fail specify all the constraints violated.

CUSTOMER

cid	cname
100	Ismail
200	Isbah
300	Tahreem
600	Izaan
800	Alia

ORDER

oid	odate	cid
1	2026-01-20	200
3	2026-01-20	600
5	2026-02-15	300
7	2026-02-20	800

PRODUCT

Pid	pname	price	company
10	Nutella	250	Ferrero
20	Kinder Joy	60	Ferrero
30	Milo	30	Nestle
40	Noodle	25	Nestle

ORDER_DETAIL

oid	pid	quantity
1	10	2
1	30	6
3	20	1
5	10	3
5	30	4

a. Insert into ORDER_DETAIL <1, 30, 7> [redacted]

b. Update oid=null in ORDER table tuple where cid=800 [redacted]

c. Delete from CUSTOMER table the tuple where cname='Alia' [redacted]

d. Delete from PRODUCT table where pid=40 [redacted]

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